

CANONICAL LESSON

Medicines Before Stock-Out

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C25
Notebook	V7_B_N05
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Anticipate medicine and vaccine stock-outs using demand, lead time, cold-chain, inventory quality, uncertainty, and governed replenishment rules.

Learning outcomes

Why this matters

Inventory totals can overstate usable supply when products are expired, inaccessible, misclassified, or cold-chain compromised.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

Worked cross-sector example

A vaccine alert combines usable stock, consumption, wastage, lead time, storage capacity, and expiry dates.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Add expiry and batch constraints.
- 2.
- Model pipeline orders.
- 3.

- Add service-level targets.
- 4.
- Explain why low stock does not justify automatic substitution.

Answer key

- 1.
- Exclude expired/soon-expiring stock and preserve batch traceability.
- 2.
- Add confirmed quantity and expected arrival with uncertainty.
- 3.
- Link safety stock to acceptable stock-out risk and criticality.
- 4.
- Substitution requires clinical/pharmacy protocols, indications, interactions, dosage forms, authorization, and communication.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.